

CENTRAL BOARD OF SECONDARY EDUCATION**HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)****SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 1: EXPLORATION: ENTERING THE WORLD OF SECONDARY SCIENCE****TIME ALLOWED: 45 MINUTES****MAXIMUM MARKS: 25****PAPER CODE: 086/CH01****Student's Name:** _____**Roll No.:** _____**Class & Section:** Class IX - _____**Date of Exam:** _____**Invigilator's Sign:** _____**GENERAL INSTRUCTIONS:**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** Standard Multiple Choice Questions with one correct option.
3. **Section B (Q16 – Q20):** Assertion-Reasoning based questions. Read both statements carefully before marking.
4. **Section C (Q21 – Q25):** Case-Study / Practical Data based questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** using a blue or black ballpoint pen.

SECTION A: MULTIPLE CHOICE QUESTIONS (Q.1 TO Q.15)**[15 MARKS]****Q.1 Which of the following is considered the fundamental base unit for temperature in the International System of Units (SI)?**

- (A) Degree Celsius (°C) (B) Kelvin (K)
(C) Fahrenheit (°F) (D) Calorie (cal)

Q.2 While measuring the volume of water in a graduated cylinder, a student notices a curved surface of the liquid. For water, the reading should be taken at:

- (A) The upper meniscus at eye level (B) The lower meniscus at eye level
(C) The midpoint between upper and lower levels (D) The upper meniscus from an angle above

Q.3 In a scientific experiment designed to test how temperature affects the rate of sugar dissolving in water, which parameter is the independent variable?

- (A) The time taken for sugar to dissolve (B) The temperature of water
(C) The amount of water used (D) The stirring speed

Q.4 Which laboratory burner flame is non-luminous, produces the highest temperature, and is used for heating without leaving soot?

- (A) A yellow flame with closed air hole (B) A blue flame with fully opened air hole
(C) An orange smoky flame (D) A red flickering flame

Q.5 A student measures the length of an object four times and gets 12.1 cm, 12.1 cm, 12.0 cm, and 12.1 cm. The actual true length is 15.0 cm. The measurements are:

- (A) Both accurate and precise (B) Accurate but not precise
(C) Precise but not accurate (D) Neither accurate nor precise

Q.6 Which of the following prefixes represents 10^{-6} in the metric measurement system?

- (A) Milli (m) (B) Micro (μ)
(C) Nano (n) (D) Pico (p)

Q.7 What is the primary purpose of having a 'control group' in a scientific controlled experiment?

- (A) To test multiple hypotheses simultaneously (B) To provide a baseline for comparison against the experimental group
(C) To ensure all variables change at once (D) To speed up the completion of the experiment

Q.8 Which piece of laboratory glassware is designed specifically for preparing solutions of exact and precise known volumes?

- (A) Beaker (B) Volumetric flask
(C) Conical flask (D) Graduated test tube

Q.9 If a balance consistently reads 0.25 g even when nothing is placed on the pan, this error is categorized as:

- (A) Random error (B) Systematic zero error
(C) Human observational error (D) Parallax error

Q.10 What safety equipment should be immediately used if a chemical splashes into a student's eyes during a science laboratory experiment?

- (A) Safety shower (B) Eyewash station for at least 15 minutes
(C) Fire blanket (D) Fume hood

Q.11 Which of the following is a derived SI unit rather than a base unit?

- (A) Second (s) (B) Kilogram (kg)
(C) Newton (N) (D) Mole (mol)

Q.12 A hypothesis in secondary science is best described as:

- (A) A proven scientific law that never changes (B) A testable and falsifiable proposed explanation for an observation
(C) An unquestionable fact accepted by scientists (D) A random guess made without background reasoning

Q.13 When heating a test tube containing a chemical liquid over a flame, the mouth of the test tube should be directed:

- (A) Towards oneself to monitor the bubbling (B) Towards the teacher only
(C) Away from oneself and all fellow classmates (D) Directly downward towards the lab bench

Q.14 The density of an aluminum block with mass 54 g and volume 20 cm³ is:

- (A) 2.7 g/cm³ (B) 0.37 g/cm³
(C) 1080 g/cm³ (D) 74 g/cm³

Q.15 In scientific notation, the diameter of a typical animal cell measuring 0.000025 m is written as:

- (A) 25×10^{-4} m (B) 2.5×10^{-5} m
(C) 0.25×10^{-6} m (D) 2.5×10^{-6} m

SECTION B: ASSERTION - REASON QUESTIONS (Q.16 TO Q.20)

[5 MARKS]

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.16 Assertion (A): SI units are accepted universally by scientists worldwide.

Reason (R): Standardized units ensure consistency, accuracy, and clear communication in scientific findings across all nations.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

Q.17 Assertion (A): Parallax error occurs when the observer's eye is placed obliquely to the measurement scale.

Reason (R): Parallax error is a type of systematic instrumental fault caused by poor calibration.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

Q.18 Assertion (A): In any controlled science experiment, all variables except the one being tested must be kept constant.

Reason (R): If multiple variables change at once, it is impossible to determine which factor caused the observed result.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

Q.19 Assertion (A): Water should never be poured directly into concentrated sulfuric acid when diluting it in a laboratory.

Reason (R): The dissolution of sulfuric acid in water is highly exothermic, and pouring water can cause boiling acid to violently splash out.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

Q.20 Assertion (A): A scientific theory can never be challenged or modified once established.

Reason (R): Science progresses as new evidence, higher precision tools, and fresh observations emerge.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

SECTION C: CASE-BASED / EXPERIMENTAL QUESTIONS (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL SETUP:

Case Study: Investigation on Rate of Evaporation

A group of Class 9 students conducted an inquiry to investigate how surface area affects the rate of evaporation of water. They took three containers: Container X (a narrow test tube, diameter 1.5 cm), Container Y (a beaker, diameter 6 cm), and Container Z (a wide petri dish, diameter 12 cm). Each container was filled with exactly 50 mL of distilled water. All three containers were placed side by side on the same laboratory table at 25°C with no wind for 24 hours. After 24 hours, the remaining water volume was measured.

Q.21 In this experiment, what is the independent variable being tested by the students?

- (A) The volume of remaining water
(B) The surface area exposed to the air
(C) The ambient room temperature
(D) The duration of 24 hours

Q.22 Which of the following serves as the dependent variable in this inquiry?

- (A) Diameter of the container
(B) Rate of evaporation / Volume of water evaporated
(C) Room temperature
(D) Type of liquid used

Q.23 Which container will show the greatest loss of water due to evaporation after 24 hours?

- (A) Container X (narrow test tube) (B) Container Y (beaker)
(C) Container Z (wide petri dish) (D) All three will lose identical amounts

Q.24 Why were all three containers kept in the exact same room at the same temperature and humidity?

- (A) To make the test unfair (B) To act as controlled variables and ensure fair comparison
(C) To increase random errors (D) Because the lab had no other rooms available

Q.25 If the students want to increase the evaporation rate in all three containers simultaneously, which modification should they make?

- (A) Increase room humidity (B) Place an electric fan nearby to increase air velocity
(C) Lower the room temperature (D) Cover the containers with glass lids

CBSE CANDIDATE OMR ANSWER RESPONSE GRID

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & DETAILED EXPLANATIONS

Q. No.	Correct	Conceptual Explanation & Syllabus Rationale
Q.1	(B)	Kelvin (K) is the SI base unit of thermodynamic temperature. Degree Celsius and Fahrenheit are derived or conventional scales.
Q.2	(B)	For liquids that wet glass like water (concave meniscus), the correct volume measurement is read at the bottom/lower meniscus at eye level to prevent parallax error.
Q.3	(B)	The independent variable is the factor deliberately altered by the experimenter (here, water temperature) to observe its effect on the dependent variable (dissolution rate).
Q.4	(B)	When the air hole of a Bunsen burner is open, complete combustion of gas occurs, producing a hot, blue, non-luminous flame without soot deposit.
Q.5	(C)	Precision refers to the closeness of repeated measurements to each other (12.0-12.1 cm), while accuracy refers to closeness to the true value (15.0 cm). Hence, precise but not accurate.
Q.6	(B)	Micro represents 10^{-6} , whereas milli is 10^{-3} , nano is 10^{-9} , and pico is 10^{-12} .
Q.7	(B)	A control group does not receive the experimental treatment and serves as a standard baseline to verify whether changes are due to the independent variable.
Q.8	(B)	A volumetric flask is calibrated to contain a precise volume of liquid at a specific temperature, ideal for standard solutions.
Q.9	(B)	A non-zero reading with zero load is a systematic zero error that shifts all measurements by a fixed amount.
Q.10	(B)	In case of chemical contact with eyes, flush immediately at an eyewash station with clean water continuously for at least 15 minutes.
Q.11	(C)	Newton ($N = \text{kg} \cdot \text{m/s}^2$) is a derived unit of force, whereas second, kilogram, and mole are fundamental base SI units.
Q.12	(B)	A scientific hypothesis must be testable through experiment and capable of being proven false (falsifiable).
Q.13	(C)	To prevent accidental injury from sudden boiling or chemical spurting, always point the open mouth of a heating test tube away from yourself and others.
Q.14	(A)	$\text{Density} = \frac{\text{Mass}}{\text{Volume}} = \frac{54 \text{ g}}{20 \text{ cm}^3} = 2.7 \text{ g/cm}^3$.
Q.15	(B)	Shifting the decimal point 5 places to the right gives $2.5 \times 10^{-5} \text{ m}$.
Q.16	(A)	The SI system was adopted globally so that scientific measurements are mutually understood and reproducible worldwide without unit confusion.
Q.17	(C)	Assertion is true (oblique viewing angle causes apparent shift). However, Reason is false because parallax error is an observational human error, not an instrumental fault.
Q.18	(A)	Controlled variables ensure a fair test by isolating the effect of the single independent variable under investigation.
Q.19	(A)	Always add acid slowly to water with constant stirring ('Acid to Water, like A to W') because water has high specific heat capacity to absorb the released heat.
Q.20	(D)	Assertion is false because science is dynamic and theories can be revised when new empirical evidence arises. Reason is true.
Q.21	(B)	The independent variable is the surface area of the water, varied by using containers of different diameters.
Q.22	(B)	The dependent variable is the volume of water evaporated, which is measured in response to changes in surface area.
Q.23	(C)	Evaporation is a surface phenomenon. Larger surface area allows more water molecules to escape into vapor phase per unit time.

Q. No.	Correct	Conceptual Explanation & Syllabus Rationale
Q.24	(B)	Keeping environmental factors like temperature, humidity, and airflow constant ensures they do not confound the effect of surface area.
Q.25	(B)	Increasing wind speed/air velocity moves vapor particles away quickly, maintaining a steep concentration gradient and accelerating evaporation.